

Computer Networks: Variable Length Subnet Mask (VLSM)

Comprehensive Study & Problem Solving Notes

August 22, 2026

In traditional **FLSM (Fixed Length Subnet Mask)**, every subnet is forced to be the exact same size regardless of how many devices are attached. This leads to massive IP address wastage (e.g., allocating 254 addresses for a point-to-point router link that only needs 2 IP addresses).

VLSM (Variable Length Subnet Mask) solves this by allowing subnets to have **different prefix lengths** based on the actual host demands of each individual network segment.

The Absolute Golden Rule of VLSM

Always sort all subnet requirements in **DESCENDING ORDER** (from LARGEST host requirement to SMALLEST host requirement) before assigning IP address ranges!

If you do not sort in descending order, larger subnets will overlap with previously allocated smaller blocks, leading to address space fragmentation and routing conflicts.

For each network segment (starting from the largest host count requirement H down to the smallest):

Step 1: Calculate Total IP Capacity Needed (H_2):

$$H_2 = H + 2 \quad (\text{Adding 1 for Network ID and 1 for Broadcast ID})$$

Step 2: Calculate Required Host Bits (H_b):

$$H_b = \lceil \log_2(H_2) \rceil = \lceil \log_2(H + 2) \rceil$$

Step 3: Calculate New CIDR Prefix Length (P):

$$P = 32 - H_b$$

Step 4: Calculate Total Block Size (IPs Allocated):

$$\text{Allocated IPs} = 2^{H_b}$$

Step 5: Determine Next Subnet Network IP:

$$\text{Next Network IP} = \text{Current Network IP} + \text{Allocated IPs}$$

Step 6: Calculate Subnet Mask:

$$\text{Subnet Mask Octet Value} = 256 - \text{Block Size (within affected octet)}$$

—

Given Base Block: 192.168.16.0/24 (Total 256 IPs available)

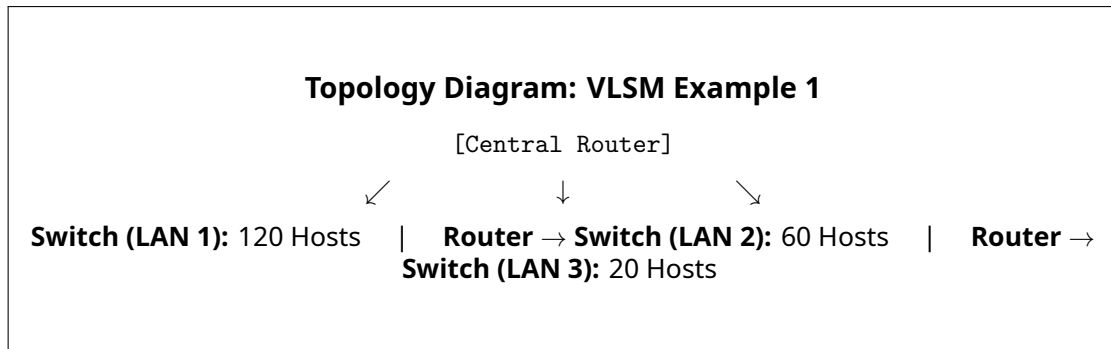


Figure 1: Network Topology for VLSM Example 1 (192.168.16.0/24).

0.1 Requirement Analysis

From the topology diagram, we have:

- **LAN 1:** 120 Hosts
- **LAN 2:** 60 Hosts
- **LAN 3:** 20 Hosts
- **WAN Link 1 (Router to Router):** 2 Hosts
- **WAN Link 2 (Router to Router):** 2 Hosts

0.2 Detailed Derivations

1. Subnet for 120 Hosts (Largest)

- $H_2 = 120 + 2 = 122$
- $H_b = \lceil \log_2(122) \rceil = 7 \text{ bits} \quad (2^6 = 64 < 122 \leq 2^7 = 128)$
- Prefix length $P = 32 - 7 = \mathbf{/25}$
- Allocated IPs $= 2^7 = \mathbf{128}$
- Network Address $= \mathbf{192.168.16.0/25}$
- Subnet Mask $= 255.255.255.(256 - 128) = \mathbf{255.255.255.128}$
- Next Network IP $= 192.168.16.0 + 128 = \mathbf{192.168.16.128}$

2. Subnet for 60 Hosts

- $H_2 = 60 + 2 = 62$
- $H_b = \lceil \log_2(62) \rceil = 6 \text{ bits} \quad (2^5 = 32 < 62 \leq 2^6 = 64)$
- Prefix length $P = 32 - 6 = \mathbf{/26}$
- Allocated IPs $= 2^6 = \mathbf{64}$
- Network Address $= \mathbf{192.168.16.128/26}$
- Subnet Mask $= 255.255.255.(256 - 64) = \mathbf{255.255.255.192}$
- Next Network IP $= 192.168.16.128 + 64 = \mathbf{192.168.16.192}$

3. Subnet for 20 Hosts

- $H_2 = 20 + 2 = 22$
- $H_b = \lceil \log_2(22) \rceil = 5 \text{ bits}$ ($2^4 = 16 < 22 \leq 2^5 = 32$)
- Prefix length $P = 32 - 5 = /27$
- Allocated IPs = $2^5 = 32$
- Network Address = **192.168.16.192/27**
- Subnet Mask = $255.255.255.(256 - 32) = 255.255.255.224$
- Next Network IP = $192.168.16.192 + 32 = 192.168.16.224$

4. Subnet for WAN Link 1 (2 Hosts)

- $H_2 = 2 + 2 = 4$
- $H_b = \lceil \log_2(4) \rceil = 2 \text{ bits}$
- Prefix length $P = 32 - 2 = /30$
- Allocated IPs = $2^2 = 4$
- Network Address = **192.168.16.224/30**
- Subnet Mask = $255.255.255.(256 - 4) = 255.255.255.252$
- Next Network IP = $192.168.16.224 + 4 = 192.168.16.228$

5. Subnet for WAN Link 2 (2 Hosts)

- $H_2 = 2 + 2 = 4 \implies H_b = 2 \text{ bits}$
- Prefix length $P = /30$, Allocated IPs = 4
- Network Address = **192.168.16.228/30**
- Subnet Mask = **255.255.255.252**

0.3 Summary Allocation Table for Example 1

Table 1: VLSM Allocation Table for 192.168.16.0/24

Subnet Name	Req. Hosts	H_b	Prefix	Network IP	Usable Host Range	Subnet Mask
LAN 1	120	7	/25	192.168.16.0	192.168.16.1 – 126	255.255.255.128
LAN 2	60	6	/26	192.168.16.128	192.168.16.129 – 190	255.255.255.192
LAN 3	20	5	/27	192.168.16.192	192.168.16.193 – 222	255.255.255.224
WAN 1	2	2	/30	192.168.16.224	192.168.16.225 – 226	255.255.255.252
WAN 2	2	2	/30	192.168.16.228	192.168.16.229 – 230	255.255.255.252

—
Given Network Address: 13.65.136.0/21

Analyzing the Base IP Block 13.65.136.0/21

- **Total Host Bits:** $32 - 21 = 11$ bits
- **Total Capacity:** $2^{11} = \mathbf{2048}$ IP addresses
- **IP Address Range:** 13.65.136.0 through 13.65.143.255
- **Default Subnet Mask:** 255.255.248.0

Campus Network Topology: VLSM Example 2 (13.65.136.0/21)

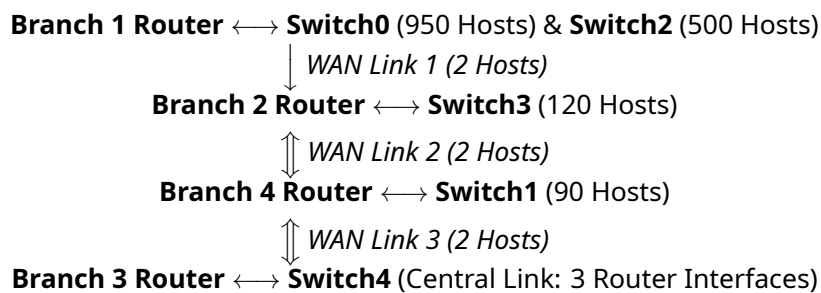


Figure 2: Multi-Branch Campus Network Topology for 13.65.136.0/21.

0.4 Topology Requirements (Sorted Descending)

Analyzing the multi-router topology diagram:

1. **Switch0 LAN:** 950 Hosts (Connected to Branch 1)
2. **Switch2 LAN:** 500 Hosts (Connected to Branch 1)
3. **Switch3 LAN:** 120 Hosts (Connected to Branch 2)
4. **Switch1 LAN:** 90 Hosts (Connected to Branch 4)
5. **Switch4 Central Link:** 3 Router Interfaces (Branch 1, Branch 3, Branch 4)
6. **WAN 1:** 2 Hosts (Branch 1 $\longleftrightarrow$ Branch 2)
7. **WAN 2:** 2 Hosts (Branch 2 $\longleftrightarrow$ Branch 4)
8. **WAN 3:** 2 Hosts (Branch 3 $\longleftrightarrow$ Branch 4)

0.5 Step-by-Step Derivations

1. Subnet 1: Switch0 LAN (950 Hosts)

- $H_2 = 950 + 2 = 952$
- $H_b = \lceil \log_2(952) \rceil = \mathbf{10}$ bits ($2^9 = 512 < 952 \leq 2^{10} = 1024$)
- Prefix length $P = 32 - 10 = \mathbf{/22}$

- Total Allocated IPs = $2^{10} = \mathbf{1024}$ IPs (Block size across 3rd octet = $1024/256 = 4$)
- Network Address: 13.65.136.0/22
- Usable Range: 13.65.136.1 to 13.65.139.254
- Broadcast Address: 13.65.139.255
- Subnet Mask: 255.255.252.0
- Next Network IP: $13.65.(136 + 4).0 = 13.65.140.0$

2. Subnet 2: Switch2 LAN (500 Hosts)

- $H_2 = 500 + 2 = 502$
- $H_b = \lceil \log_2(502) \rceil = \mathbf{9}$ bits ($2^8 = 256 < 502 \leq 2^9 = 512$)
- Prefix length $P = 32 - 9 = \mathbf{/23}$
- Total Allocated IPs = $2^9 = \mathbf{512}$ IPs (Block size across 3rd octet = $512/256 = 2$)
- Network Address: 13.65.140.0/23
- Usable Range: 13.65.140.1 to 13.65.141.254
- Broadcast Address: 13.65.141.255
- Subnet Mask: 255.255.254.0
- Next Network IP: $13.65.(140 + 2).0 = 13.65.142.0$

3. Subnet 3: Switch3 LAN (120 Hosts)

- $H_2 = 120 + 2 = 122 \implies H_b = \mathbf{7}$ bits ($2^7 = 128$)
- Prefix length $P = 32 - 7 = \mathbf{/25}$
- Total Allocated IPs = $2^7 = \mathbf{128}$ IPs
- Network Address: 13.65.142.0/25
- Usable Range: 13.65.142.1 to 13.65.142.126
- Broadcast Address: 13.65.142.127
- Subnet Mask: 255.255.255.128
- Next Network IP: $13.65.142.0 + 128 = 13.65.142.128$

4. Subnet 4: Switch1 LAN (90 Hosts)

- $H_2 = 90 + 2 = 92 \implies H_b = \mathbf{7}$ bits ($2^7 = 128$)
- Prefix length $P = 32 - 7 = \mathbf{/25}$
- Total Allocated IPs = $2^7 = \mathbf{128}$ IPs
- Network Address: 13.65.142.128/25
- Usable Range: 13.65.142.129 to 13.65.142.254
- Broadcast Address: 13.65.142.255
- Subnet Mask: 255.255.255.128
- Next Network IP: $13.65.142.128 + 128 = 13.65.143.0$

5. Subnet 5: Switch4 Central Link (3 Router Interfaces)

- $H_2 = 3 + 2 = 5 \implies H_b = \mathbf{3}$ bits ($2^2 = 4 < 5 \leq 2^3 = 8$)
- Prefix length $P = 32 - 3 = \mathbf{/29}$
- Total Allocated IPs = $2^3 = \mathbf{8}$ IPs
- Network Address: 13.65.143.0/29
- Usable Range: 13.65.143.1 to 13.65.143.6
- Broadcast Address: 13.65.143.7
- Subnet Mask: 255.255.255.248
- Next Network IP: $13.65.143.0 + 8 = 13.65.143.8$

6. Subnet 6: WAN Link 1 (Branch 1 $\longleftrightarrow$ Branch 2)

- $H_2 = 2 + 2 = 4 \implies H_b = \mathbf{2}$ bits ($2^2 = 4$)
- Prefix length $P = 32 - 2 = \mathbf{/30}$, Total IPs = **4**
- Network Address: 13.65.143.8/30
- Usable Range: 13.65.143.9 to 13.65.143.10
- Subnet Mask: 255.255.255.252
- Next Network IP: $13.65.143.8 + 4 = 13.65.143.12$

7. Subnet 7: WAN Link 2 (Branch 2 $\longleftrightarrow$ Branch 4)

- $H_2 = 2 + 2 = 4 \implies H_b = \mathbf{2}$ bits, $P = \mathbf{/30}$
- Network Address: 13.65.143.12/30
- Usable Range: 13.65.143.13 to 13.65.143.14
- Subnet Mask: 255.255.255.252
- Next Network IP: $13.65.143.12 + 4 = 13.65.143.16$

8. Subnet 8: WAN Link 3 (Branch 3 ↔ Branch 4)

- $H_2 = 2 + 2 = 4 \implies H_b = 2$ bits, $P = /30$
- Network Address: 13.65.143.16/30
- Usable Range: 13.65.143.17 to 13.65.143.18
- Subnet Mask: 255.255.255.252

Table 2: Complete Master VLSM Allocation Table for 13.65.136.0/21

Subnet	Hosts	Prefix	Network IP	First Usable IP	Last Usable IP	Subnet Mask
Switch0	950	/22	13.65.136.0	13.65.136.1	13.65.139.254	255.255.252.0
Switch2	500	/23	13.65.140.0	13.65.140.1	13.65.141.254	255.255.254.0
Switch3	120	/25	13.65.142.0	13.65.142.1	13.65.142.126	255.255.255.128
Switch1	90	/25	13.65.142.128	13.65.142.129	13.65.142.254	255.255.255.128
Switch4	3	/29	13.65.143.0	13.65.143.1	13.65.143.6	255.255.255.248
WAN 1	2	/30	13.65.143.8	13.65.143.9	13.65.143.10	255.255.255.252
WAN 2	2	/30	13.65.143.12	13.65.143.13	13.65.143.14	255.255.255.252
WAN 3	2	/30	13.65.143.16	13.65.143.17	13.65.143.18	255.255.255.252